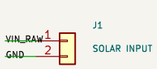
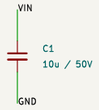
1. SOLAR INPUT / ENABLE

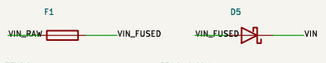


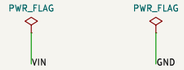




Panel: 12 V nominal, Vmp ~18 V
Cold Voc <=25 V. F1/D5 protect solar input.
Battery reversal and overvoltage are NOT protected.

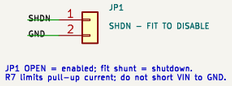
FUSED REVERSE-POLARITY SOLAR INPUT







2. LT3652 CHARGER / POWER STAGE

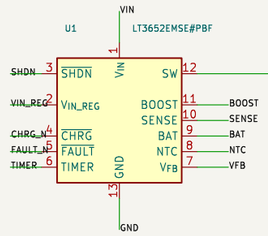


JP1 OPEN = enabled; fit shunt = shutdown.
R7 limits pull-up current; do not short VIN to GND.

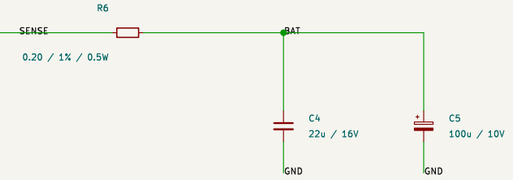












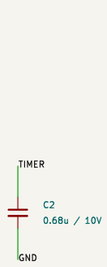
R6 sets 500 mA maximum charge current.
1S protected Li-Ion cell only: 4.20 V full charge.
Battery connector pin 1 = BAT+; pin 2 = GND.

3. PANEL VOLTAGE REGULATION



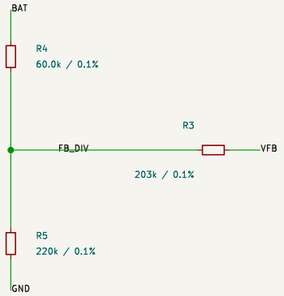
Protected VIN regulation target = 17.874 V.
Panel target = 17.874 V + D5 drop.
Fixed voltage regulation; not dynamic MPPT.

4. TIMER



Nominal 3 h; bad-battery
preconditioning timeout ~22.5 min.
Timer retries if float not reached.

5. PRECISION BATTERY FEEDBACK



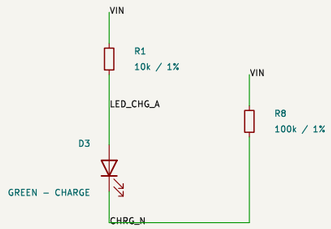
Vfloat = 3.3*(1+60.0k/220k) = 4.200 V
R3 + (R4 || R5) = 250.14 kohm
Use specified 0.1% feedback resistors.

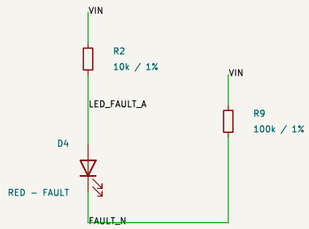
6. TEMPERATURE SENSOR



Attach 10k NTC, B=3380 K, to cell.
Nominal charge window ~0 to 40 C.
Open sensor DISABLES temperature monitoring.

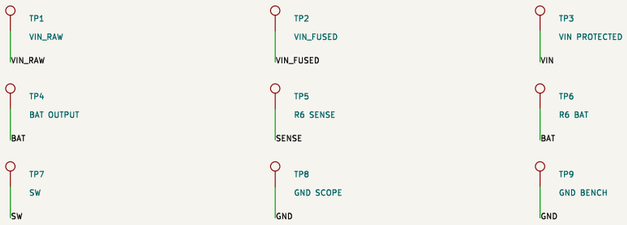
7. ADDED STATUS LEDs





Green only: charging >=C/10 (50 mA nominal).
Red only: bad-battery timeout. Both: NTC fault.
Both off: standby, no input, or low charging current.
LEDs are powered from solar VIN; off does not prove full charge.

8. REV B PROBE POINTS (see test plan)





VIN-LEVEL STATUS
NOT MCU SAFE
VIN_REG: analog sense only

ENGINEERING PROTOTYPE - ERC/DRC do not replace bench verification. No lithium cell until dummy-load checks pass.

Based on the uploaded Fernando S.M.H.G. project; corrected circuit and layout
1S Li-ion / 4.20 V / 500 mA - engineering prototype

Sheet: /
File: Solar_Charger_LED_RevB.kicad_sch

Title: Solar Charger Rev B - Fused Input, Shutdown and Test Points

Size: A3	Date: 2026-11-06	**Rev: B**
KiCad E.D.A. 9.0.2		Id: 1/1